

Sensor Systems

Laboratory 0 5
USART

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1. USART

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USART

Universal Synchronous and Asynchronous Receiver - Transmitter

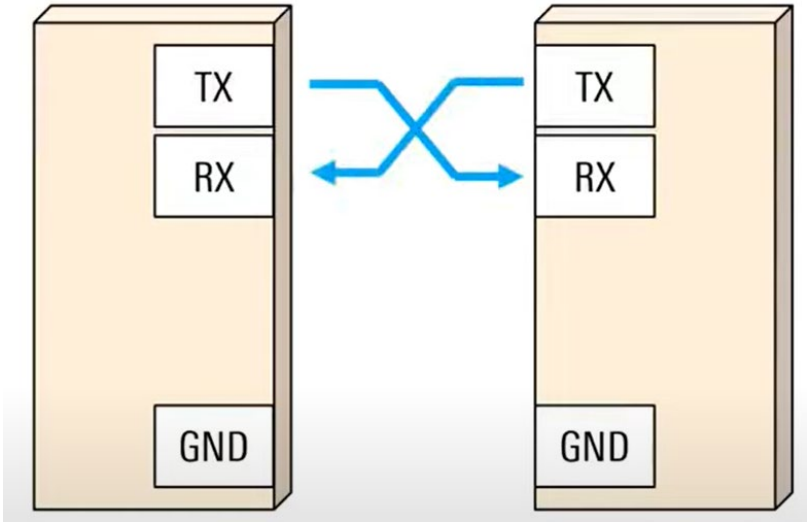
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type of a serial interface device that can be programmed to communicate asynchronously or synchronously.

- COM ports provide a convenient way for **PCs and embedded systems to exchange information** .
- The traditional COM port on a PC is an RS-232 serial port. Recent PCs often skip RS-232 in favor of USB. Nevertheless a **USB device** can appear as a **virtual COM** port that applications can access using COM-port APIs or libraries.
- Many existing devices with asynchronous serial ports can use a USB UART to communicate with PCs as a virtual COM.

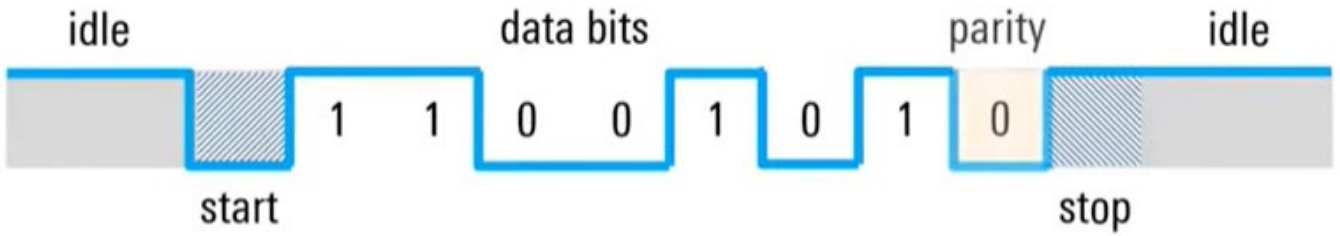
What is UART?

- Protocol for **exchanging serial data between two devices** .
- Uses only **2 wires** : **TX → RX**, in each direction, plus a common ground.
- Transmitter and receiver **do not share a common** clock: they must agree on the speed (**Baud rate**) and **parameters of the transmission** .



Data is transmitted as "frames":

- **Start/stop** bits
- **Data** bits (5 to 9, usually 7 or 8, sent LSB first)
- **Parity** bit (optional) useful for error detection:
  Even parity: '1' if number of '1' is even / **Odd** parity: '1' if number of '1' is odd.



2. Functions and exercises

02

UART HAL functions

Basics functions:

HAL_StatusTypeDef **HAL_UART_Receive** (UART_HandleTypeDef *huart , uint8_t * pData ,
uint16_t Size, uint32_t Timeout)

HAL_StatusTypeDef **HAL_UART_Transmit** (UART_HandleTypeDef *huart , uint8_t * pData ,
uint16_t Size, uint32_t Timeout)

Direct Memory Access functions:

HAL_StatusTypeDef **HAL_UART_Receive_DMA** (UART_HandleTypeDef *huart , uint8_t * pData ,
uint16_t Size)

HAL_StatusTypeDef **HAL_UART_Transmit_DMA** (UART_HandleTypeDef *huart , uint8_t * pData ,
uint16_t Size)

Add _IT for the interrupt version!

UART HAL functions

Function to receive an amount of data in blocking mode till either the **expected number of data is received or an IDLE event occurs.**

HAL_StatusTypeDef **HAL_UARTEx_ReceiveToIdle** (UART_HandleTypeDef *huart , uint8_t * pData , uint16_t Size, uint16_t * RxLen, uint32_t Timeout)

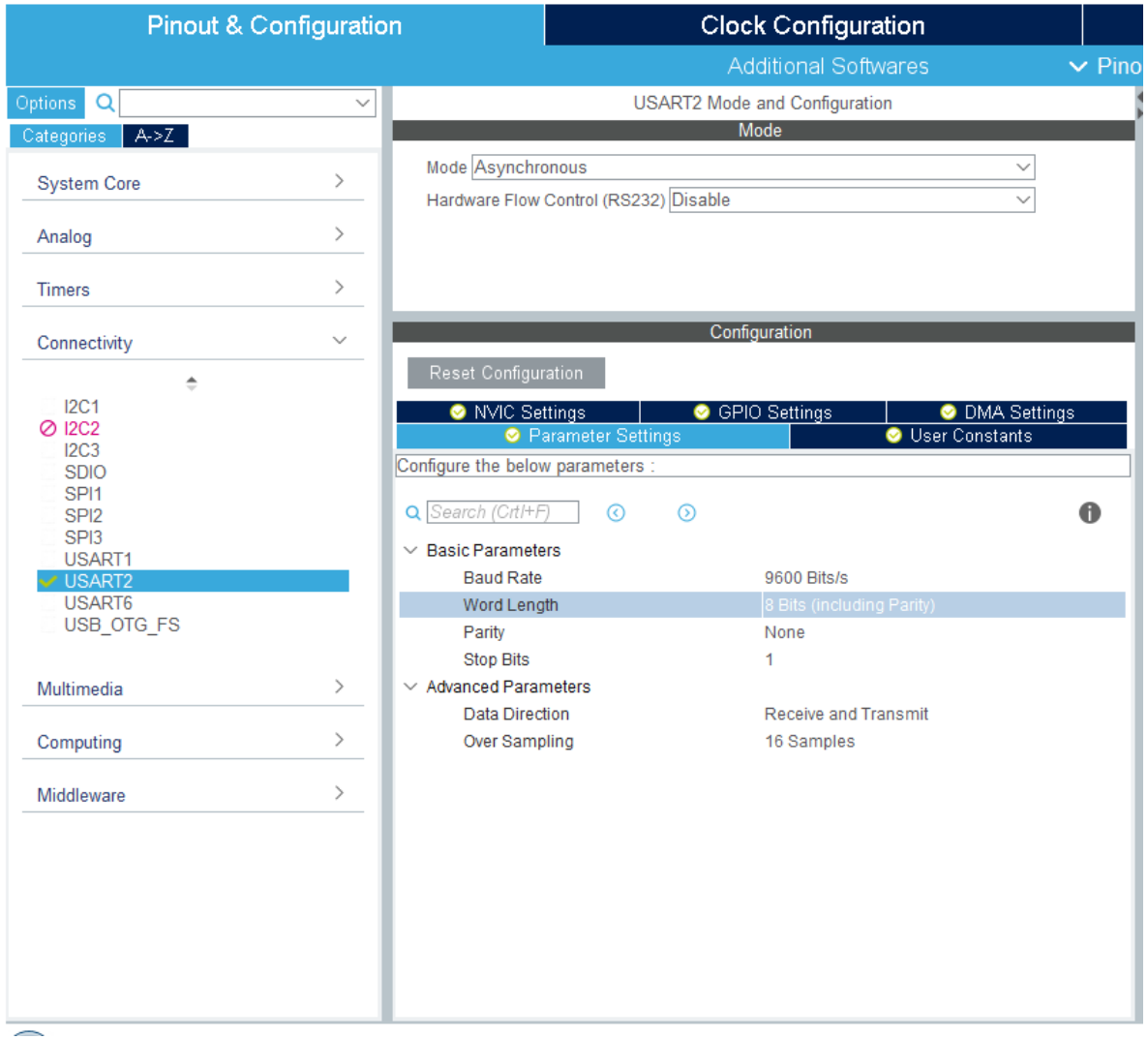
HAL_StatusTypeDef **HAL_UARTEx_ReceiveToIdle_DMA** (UART_HandleTypeDef *huart , uint8_t * pData , uint16_t Size)

Project 3a

The objective of this project is to **send information from the microcontroller to the PC**, using the USART interface for the Virtual COM.

- You will send a string containing your name and your year of birth followed by a new line every second.
- We will use a terminal emulator (Arduino IDE, VScode, Putty) to receive the string.

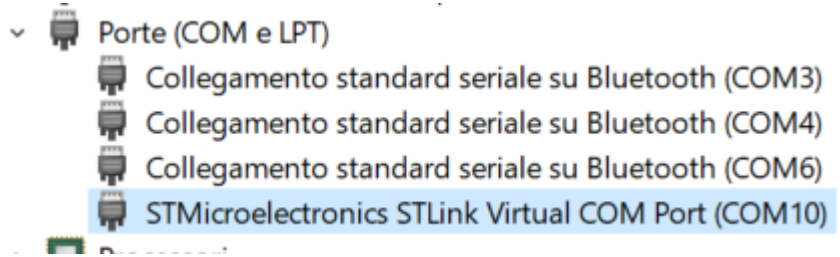
USART configuration



- Open a new project (use default)
- Check that the two pins for USART are enabled

- Select a Baud Rate (e.g. 115200 Bits/s). The same BR should be set in the terminal emulator, since it is an asynchronous communication!

Identify the COM port on your PC (Windows users)



- Open "Device Manager" (Gestione dispositivi) by right-clicking on the start menu, and selecting "Device Manager"
- Scroll down until "Ports (COM and LPT)" and open the submenu. The COM port we need should be identified by the name "STMicroelectronics STLink Virtual COM Port (COM__)"

Generate string

To correctly create the data to be sent to the PC, use the **snprintf** function to generate a string.

Example:

```
uint8_t count = 10;
int length = snprintf (string, sizeof (string), " %d\n", count);
HAL_UART_Transmit (&huart2, string, length, 100);
```

Example:

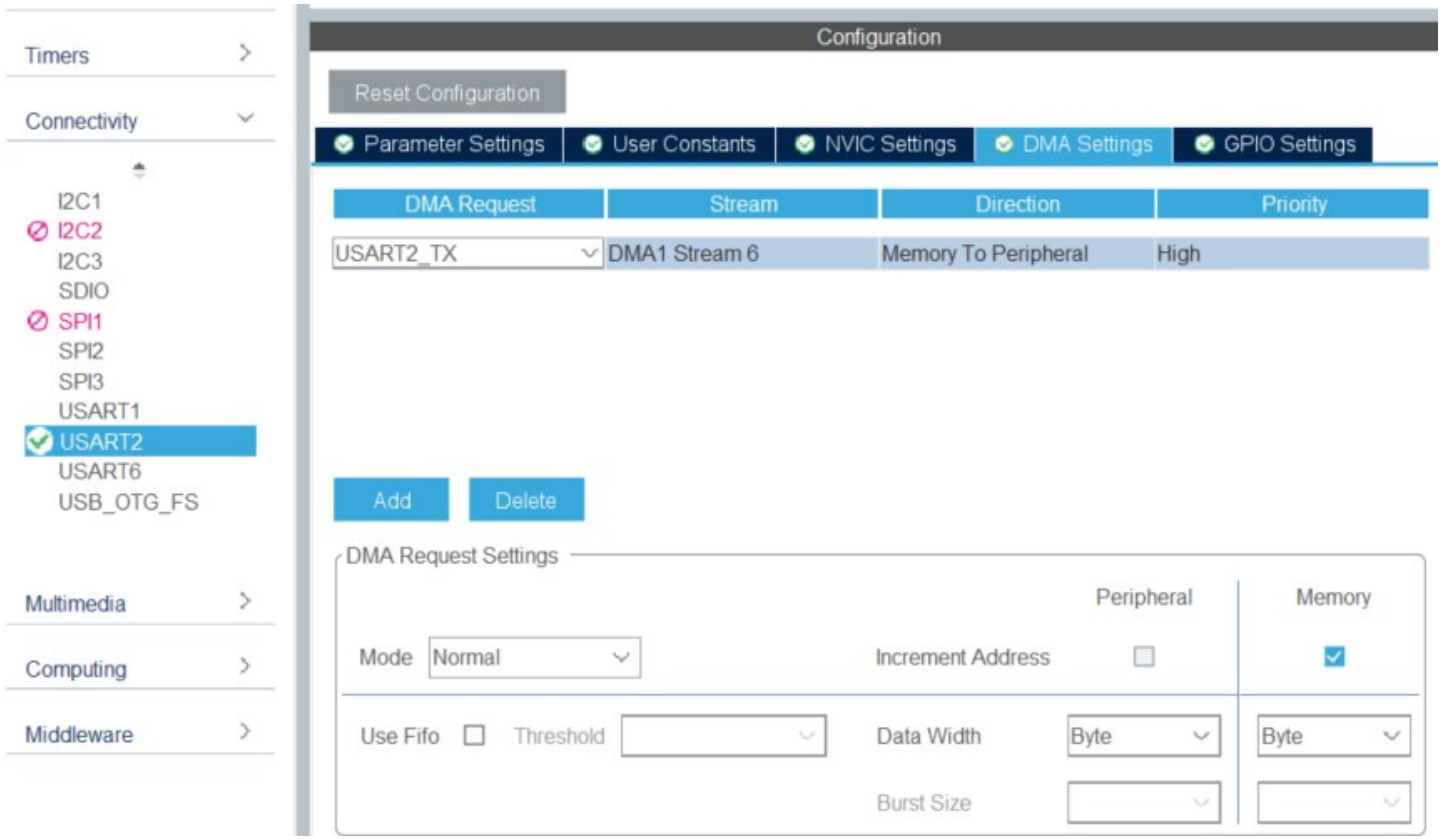
```
float voltage =0.125;
float voltage2 = 10.535;
int length = snprintf (string, sizeof (string), "%.3f,%.3f \n", voltage,voltage2);
HAL_UART_Transmit (&huart2, string, length, 100);
```

HOMEWORK 4a

The objective of this project is to **send information from the microcontroller to the PC**, using the USART interface for the Virtual COM and **using DMA functions.**

- You will send a string containing your name and your year of birth followed by a new line every second.
- We will use a terminal emulator (Arduino IDE, VScode, Putty) to receive the string.

UART configuration



- Enable DMA in the USART2 peripheral with USART2_TX requests, in normal mode
- Enable both the interrupt for UART2 and the one for DMA.

HOMEWORK 4b

Try sending **from the PC via UART** a **string** of **variable length** .
When you send the string "LED ON", turn the green LED on.
When you send the string "LED OFF", turn the green LED off.

Hint: Use the ReceiveToIdle function.

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